

GUARULHOS, Brazil

WMO: 830750

Lat: 23.432S

Lon: 46.470W

Elev: 2459

StdP: 13.44

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	46.3	48.6	40.7	41.0	61.1	44.2	46.9	59.8	17.3	70.6	15.7	68.7	3.4	80	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.0	91.2	70.9	89.2	70.8	87.4	70.5	76.0	83.4	74.9	82.4	73.8	81.3	9.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.8	138.3	77.6	73.2	135.3	77.3	71.8	128.7	76.3	41.4	83.7	40.3	83.9	39.2	79.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.9	12.2	DB	40.5	94.4	3.3	1.8	38.1	95.7	36.2	96.7	34.3	97.7	31.9	99.0	(4)
(5)			WB	40.1	78.2	3.4	2.5	37.6	80.0	35.6	81.5	33.7	82.9	31.2	84.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	74.6	75.2	73.5	70.7	65.3	63.4	62.8	65.0	67.3	69.9	70.6	73.6	(6)	
	DBStd	6.27	4.18	3.67	3.65	4.14	4.41	4.53	4.87	5.12	5.60	5.44	4.86	4.13	(7)	
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD65	370	1	0	1	7	51	80	97	63	40	18	11	1	(9)	
	CDD50	7045	763	705	730	620	475	403	397	464	520	618	618	732	(10)	
	CDD65	1939	299	285	265	177	61	32	28	62	110	172	180	268	(11)	
	CDH74	11683	1710	1745	1335	882	296	214	296	645	900	1097	1008	1555	(12)	
	CDH80	3653	592	623	400	202	35	10	27	161	344	430	310	519	(13)	
(14) Wind	WSAvg	5.9	5.9	5.8	5.6	5.5	5.4	5.4	5.5	5.8	6.4	6.4	6.5	6.3	(14)	
(15) Precipitation	PrecAvg	67.9	11.6	8.9	8.1	4.1	3.1	2.8	2.7	1.7	4.1	5.4	6.6	8.7	(15)	
	PrecMax	94.6	23.5	16.2	15.7	7.6	7.4	10.5	7.4	3.9	8.9	8.5	12.2	15.7	(16)	
	PrecMin	50.8	6.5	4.3	3.5	0.9	0.8	0.4	0.7	0.6	1.3	1.7	3.0	3.4	(17)	
	PrecStd	10.0	4.0	3.0	2.4	1.6	1.3	2.2	1.8	0.9	2.0	1.7	2.2	2.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.8	91.8	89.7	87.6	82.8	81.0	82.7	87.8	91.8	93.3	90.0	91.5	(19)	
		MCWB	73.7	72.4	72.3	71.0	67.7	64.8	63.5	64.9	67.3	69.2	69.9	72.9	(20)	
	2%	DB	89.5	89.6	87.4	84.6	80.6	78.9	80.4	84.3	88.0	89.4	86.4	88.1	(21)	
		MCWB	73.1	72.4	71.8	70.2	66.7	64.2	62.4	63.6	65.1	68.6	69.7	71.7	(22)	
	5%	DB	86.3	87.4	84.4	82.4	77.3	76.8	77.3	80.9	84.2	85.6	84.0	85.9	(23)	
		MCWB	72.6	72.0	71.8	69.7	65.3	63.4	62.3	62.6	64.9	68.7	70.0	71.4	(24)	
	10%	DB	82.8	84.2	82.1	79.2	73.7	73.4	74.5	77.3	79.1	80.5	80.4	82.5	(25)	
		MCWB	72.3	72.3	71.4	68.7	64.5	62.9	61.7	62.2	64.9	68.2	69.0	71.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.3	77.5	76.1	74.7	70.9	68.9	66.5	68.3	71.3	73.6	74.5	76.9	(27)	
		MCDB	85.1	83.9	83.7	81.1	77.9	76.6	75.9	79.6	84.0	85.5	83.7	84.0	(28)	
	2%	WB	75.8	75.6	74.5	72.8	69.0	66.5	65.3	66.3	69.0	71.6	72.6	74.9	(29)	
		MCDB	83.1	82.4	82.0	79.9	75.5	73.4	74.5	77.2	79.4	81.6	81.5	83.3	(30)	
	5%	WB	74.6	74.5	73.4	71.5	67.6	65.3	63.8	65.0	67.5	70.2	71.3	73.6	(31)	
		MCDB	81.6	81.5	80.5	78.2	73.8	72.0	72.7	75.3	77.2	79.4	79.7	81.7	(32)	
	10%	WB	73.4	73.4	72.2	70.3	66.1	63.8	62.7	63.5	65.9	68.9	70.1	72.2	(33)	
		MCDB	80.4	80.4	78.5	76.3	71.4	70.3	71.0	72.7	74.7	76.9	76.9	79.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	17.0	16.0	16.5	16.3	18.0	19.3	21.1	18.9	17.6	16.6	17.0	(35)	
		MCDBR	20.8	21.9	20.2	20.4	21.0	23.1	25.7	28.6	28.8	26.2	23.1	22.0	(36)	
	5% WB	MCWBR	7.7	7.6	7.6	7.9	8.6	9.9	10.6	10.6	9.9	9.0	8.8	8.4	(37)	
		MCWBR	18.2	18.7	17.9	17.6	18.0	19.2	21.2	23.1	22.1	20.9	19.0	18.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.395	0.394	0.383	0.367	0.348	0.349	0.359	0.394	0.486	0.449	0.421	0.403	(40)	
		taud	2.459	2.444	2.479	2.495	2.522	2.502	2.457	2.329	2.057	2.232	2.324	2.410	(41)	
	Ebn at Noon		301	297	292	283	275	268	269	269	257	278	292	299	(42)	
		Edh at Noon	38	38	35	32	29	28	30	37	53	47	43	40	(43)	
(44) All-Sky Solar Radiation	RadAvg		1648	1746	1546	1401	1129	1041	1095	1321	1405	1569	1652	1731	(44)	
	RadStd		239	155	152	117	85	100	114	115	179	179	97	122	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.54	-0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page